DRIFTLESS — Technology Stack & Dependencies Analysis

1. Complete Technology Inventory

Technology	Category	Actual Usage	File / Module	Purpose	Related Features
Python 3.10+	Programming Language	Core Backend Execution	`dashboard/**/*.py`, `scripts/*.py`	Primary language executing web server, models, serializers, and tooling	All backend APIs, migrations, seeding, extension packaging
JavaScript (ES2022)	Programming Language	Client & Extension Execution	`extension/*.js`, `<script>` in HTML	Client-side reactive logic, timer tickers, event dispatching, REST fetches	Active tab tracking, chart rendering, theme switching, sync
HTML5	Markup Language	UI Document Structure	`dashboard/core/templates/*.html`, `extension/*.html`	Semantic layout, forms, accessible widgets, dialog overlays	Analytics views, login forms, extension popup, blocked landing
CSS3 & CSS Custom Props	Styling System	Visual Tokens & Layout	`dashboard/ … /theme.css`, `extension/theme.css`	Design system variables (`--pri`, `--bg`, `--ink`, etc.), micro-transitions	Calm Sky (Light) and Opal Noir (Dark) dual-theme system
Django 4.2 LTS	Web Framework	Server Architecture	`dashboard/driftless/*`, `dashboard/core/*`	MVC architecture, routing, ORM, authentication backend, template engine	User management, session auth, admin console, static collection
Django REST Framework (DRF 3.15)	REST API Toolkit	Web API Layer	`dashboard/core/serializers.py`, `dashboard/core/views.py`	Serializer validation, token authentication, rate limiting, JSON responses	`/api/usage/`, `/api/summary/`, `/api/limits/`, `/api/settings/`

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Chrome Manifest V3	Extension Platform Spec	Browser Extension Runtime	`extension/manifest.json`, `extension/background.js`	Defines permissions, service worker lifecycle, alarms, host matches	Browser integration, tab tracking, background execution
declarativeNetRequest API	Chrome Browser API	Network Interception	`extension/background.js`	Dynamic URL redirect rules executed at network layer without content scripts	3-tier escalation blocking, Focus Mode distraction suppression
chrome.storage.local	Browser Storage API	Local Extension State	`extension/background.js`, `extension/popup.js`	High-speed local persistence for accumulated seconds and sync queues	Offline dwell tracking, uncommitted delta queue, cached settings
chrome.alarms API	Chrome Browser API	Scheduled Background Execution	`extension/background.js`	Periodic timer triggers (`ALARM_PERIODIC_SYNC`, `ALARM_MIDNIGHT_RESET`)	60-second delta sync, daily midnight usage resets
chrome.idle API	Chrome Browser API	State Detection	`extension/background.js`	Detects when user becomes inactive for > 60 seconds	Suppresses false positive time accumulation while user is away
chrome.notifications API	Chrome Browser API	System Desktop Notifications	`extension/background.js`	Displays OS-level desktop notification cards	Tier 1 (80% Nudge), Tier 2 (100% Alert), Focus complete alert
SQLite 3	Relational Database	Data Persistence	`dashboard/db.sqlite3`	Zero-configuration file-based SQL database storage	Storage of users, profiles, usage logs, limits, and settings
Chart.js 4.4.3 (UMD)	Data Visualization	Client Charting Engine	`dashboard/core/templates/dashboard.html`	Canvas-based rendering of 7-day multi-dataset stacked bar & line charts	7-day category breakdowns, hover tooltips, mode switching
Tailwind CSS 3.x (CDN)	Utility CSS Engine	Rapid UI Component Layout	`dashboard/core/templates/*.html`	Utility classes for flex, grid, spacing, responsive visibility	Dashboard sidebar, grid cards, modal dialogs, status badges
WhiteNoise 6.6	Static Asset Serving	Middleware File Server	`dashboard/driftless/settings.py`	Serves static assets directly from Gunicorn/WSGI with compression	Production serving of CSS, images, and brand logo

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django-cors-headers 4.3	Security Middleware	CORS Policy Enforcement	`dashboard/driftless/settings.py`	Grants cross-origin request permissions to `chrome-extension://*`	Extension communication with Django REST endpoints
python-dotenv 1.0	Configuration Utility	Environment Variable Loader	`dashboard/driftless/settings.py`	Parses `.env` configuration file into `os.environ`	Environment secrets, debug flags, allowed hosts, CORS flags
Gunicorn 21.2	WSGI HTTP Server	Production Web Server	`dashboard/Procfile`	Production-grade multi-worker WSGI HTTP server	Production PaaS deployment (Render / Heroku)
Google Fonts (Inter)	Web Typography	Font Delivery	`dashboard/core/templates/*.html`	Modern geometric sans-serif typeface for clean UI legibility	Typography across dashboard and authentication pages

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2. In-Depth Component Analysis

2.1. Python & Django 4.2 LTS

- **What It Is:** High-level Python web framework providing an ORM, request router, authentication middleware, and administrative interface.
- **Why It Is Used:** Enables rapid development of secure, maintainable backend applications with robust schema migration tools and native SQL injection protection.
- **Usage Location:** All files in `dashboard/driftless/` and `dashboard/core/`.
- **Key Features Depending On It:** User registration, session login, relational database transactions, static file orchestration, template rendering.
- **Interactions:** Serves as the foundation hosting Django REST Framework and executing all database queries against SQLite.
- **Role:** Directly used core framework.

2.2. Django REST Framework (DRF)

- **What It Is:** Powerful, flexible toolkit for building Web APIs in Django.
- **Why It Is Used:** Standardizes token-based authentication (`TokenAuthentication`), incoming payload validation, rate-limiting (`UserRateThrottle`), and structured JSON serialization.
- **Usage Location:** `dashboard/core/serializers.py` , `dashboard/core/views.py` .

- **Key Features Depending On It:** All endpoints under `/api/`, including `/api/usage/`, `/api/summary/`, `/api/limits/`, `/api/settings/`, and `/api/sessions/`.
- **Interactions:** Bridges the client-side JavaScript `fetch()` calls from both the dashboard and the Chrome extension to the Django ORM.
- **Role:** Directly used core library.

2.3. Chrome Extension Manifest V3 Runtime

- **What It Is:** The modern standard for Google Chrome extensions requiring service workers instead of persistent background pages.
- **Why It Is Used:** Mandatory requirement for modern Chrome extensions; ensures memory efficiency and enhanced browser security.
- **Usage Location:** `extension/manifest.json`, `extension/background.js`, `extension/popup.js`.
- **Key Features Depending On It:** Dynamic tab listening (`tabs.onActivated`, `tabs.onUpdated`), window focus detection (`windows.onFocusChanged`), idle monitoring, and dynamic declarative network requests.
- **Interactions:** Communicates with the Django REST API over HTTP using the native browser `fetch` API.
- **Role:** Directly used execution platform.

2.4. Chart.js 4.4.3

- **What It Is:** Open-source HTML5 Canvas charting library.
- **Why It Is Used:** Renders high-performance, fluid, responsive bar and line charts with custom tooltip styling and smooth theme transitions.
- **Usage Location:** `dashboard/core/templates/dashboard.html` (`renderChart()` function).
- **Key Features Depending On It:** 7-Day category breakdown chart displaying Productivity, Social, Entertainment, and Other categories.
- **Interactions:** Dynamically reads CSS theme variables (`--prod`, `--soc`, `--ent`, `--tooltip-bg`) directly from `window.getComputedStyle()` so chart colors update instantaneously when toggling between Calm Sky and Opal Noir.
- **Role:** Directly used frontend library loaded via CDN.

2.5. Tailwind CSS & theme.css Dual-Theme Bridge

- **What It Is:** A hybrid styling approach pairing utility classes with structured CSS custom properties.
- **Why It Is Used:** Tailwind handles responsive layouts, flexboxes, and padding, while `theme.css` manages design tokens (colors, radii, shadows, borders) for Calm Sky (Light) and Opal Noir (Dark).
- **Usage Location:** `dashboard/core/templates/*.html`, `dashboard/core/static/css/theme.css`, `extension/theme.css`.

- **Key Features Depending On It:** Zero-FOUC theme rendering, high-contrast dark mode, and responsive layout scaling across desktop and mobile.
- **Role:** Directly used styling system.
